APPENDIX D

Measurement Invariance (MICOM)

Table D1

Compositional Invariance of the Addiction Composite Across Group Comparisons

Comparison	*c*	5% *q*.	*p*	*p* means	*p* variances	Invariance
TikTok vs. Instagram	.956	.947	.084	.605	.626	Full
TikTok vs. YouTube	.963	.957	.081	< .001	.126	Partial
Instagram vs. YouTube	.966	.959	.084	< .001	.077	Partial
Male vs. Female	.995	.973	.717	.017	.003	Partial
Two-parent vs. Single-parent	.974	.898	.539	.006	.216	Partial
Without vs. With older siblings	.959	.977	.004	.963	.043	Not established
ESO vs. Bachillerato	.674	.862	.001	< .001	.024	Not established

Note. MICOM procedure (Henseler et al., 2016), 5,000 permutations. *c* = compositional correlation of the formative Addiction composite; 5% *q*. = 5% quantile of its permutation distribution; *p* means and *p* variances = permutation *p*-values for the equality of composite means and variances. Full = compositional invariance plus equal means and variances; Partial = compositional invariance only. EIDAN satisfied compositional invariance in all seven comparisons (smallest $p = .197$, TikTok vs. YouTube); sleep quality and self-esteem are single-item composites and satisfy it by construction.

References

Henseler, J., Ringle, C. M., & Sarstedt, M. (2016). Testing measurement invariance of composites using partial least squares. International marketing review, 33(3), 405-431.